

Lie Groups for State Estimation in Robotics

deepseek-v4.1-flash

A robot's pose is not a list of numbers: orientations live on a curved set, so averaging, subtracting, and nudging them with ordinary vector arithmetic leave that set. This note asks nine questions that rebuild state estimation on it. What does the constraint on a rotation do to the geometry of the headings? What object captures the directions a rotation can move in? How does the exponential map turn a tangent vector back into a rotation, and how do the plus and minus operators replace addition and subtraction? When the same increment is written in two frames, what map relates them? What replaces the derivative and the chain rule on a curved state, and where does a covariance live? How does an ordinary Kalman filter change once its state is a pose? And what changes when the state carries a second, unrelated block? All nine are posed against one wheeled robot localizing in a plane, so an answer can be checked against the robot rather than against this page.

October 01, 2026

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1. Why a heading is not a number

Question: The estimator needs three ordinary vector operations on the robot's heading: average two estimates, subtract one from another to form a correction, and add a small step to move it. Which of these operations fails to land on a heading, and what property of the set of rotations – the one visible in the constraint $R^T R = I$ – is responsible? Predict where the average of two nearby headings ends up.

The running example. A wheeled robot drives in a plane and estimates its pose

$$M = \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} \in \text{SE}(2), \quad R \in \text{SO}(2), \quad t \in \mathbb{R}^2, \quad R^T R = I, \quad \det R = 1, \quad (1)$$

with orientation R and position t . It measures range and bearing to a few beacons whose positions are known, and its estimator keeps a mean and a covariance for the pose. Orientation is the part that is not a vector: in the plane $R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ is fixed by one angle θ , and in three dimensions by a rotation matrix with the same constraint. Every section below returns to this robot.

Facts to reason with.

- $R^T R = I$ says exactly that R is orthogonal, and $\det R = 1$ selects the rotations from the reflections.
- Products and inverses of rotations are rotations, and the identity is a rotation, so the set is closed under the operations an estimator composes.
- Headings are periodic: θ and $\theta + 2\pi$ name the same rotation, so a heading has no unique numerical value.
- The standard filter stores an estimate as a vector, adds a correction vector to it, and averages measurements with a weighted sum.
- The set of planar headings is the unit circle; the set of 3D rotations cannot be drawn, but the same reasoning applies to it.

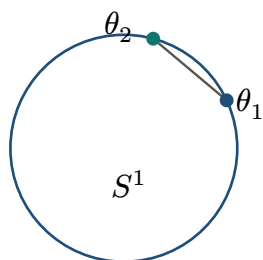


Figure 1: Two heading estimates on the unit circle, joined by the straight chord. Inspect where the chord's midpoint falls, and where a weighted sum of the two headings falls, relative to the circle.

2. The directions a rotation can move in

Question: The constraint $R^T R = I$ has to hold at every instant. Differentiating it along a path $R(t)$ with $R(0) = I$ forces the velocity into a particular family of matrices. Why is that family a genuine vector space – closed under sums and scalar multiples – when the rotations themselves are not, and what does one element of it describe about the robot?

Setup. Let $R(t)$ be a smooth path of rotations with $R(0) = I$ and $\dot{R} = dR/dt$. Differentiating $R^T R = I$ and setting $t = 0$ leaves $\dot{R} + \dot{R}^T = 0$, so the velocity at the identity is a skew-symmetric matrix. The set of such matrices is the Lie algebra of the group, written $\mathfrak{m}$, and it is the tangent space $T_{\mathcal{E}}\mathcal{M}$ at the identity. In the plane $\mathfrak{m}$ is one-dimensional; in three dimensions it is the space of skew-symmetric 3×3 matrices, in one-to-one correspondence with vectors $\omega \in \mathbb{R}^3$:

$$[\omega]_{\times} = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}, \quad \omega = ([\omega]_{\times})^{\vee}, \quad (2)$$

the map $\omega \mapsto [\omega]_{\times}$ being the **hat** operator and $[\omega]_{\times} \mapsto \omega$ the **vee** operator.

Facts to reason with.

- A skew-symmetric 3×3 matrix has three independent entries, which is why three numbers describe it; the planar case has one.
- The robot's angular velocity is the element of $\mathfrak{m}$ that describes its instantaneous turning.
- The construction works at any rotation X , not just the identity: $T_X\mathcal{M}$ is the tangent space at X .

3. Wrapping a tangent vector back onto the group

Question: For a constant angular velocity the path obeys $\dot{R} = R[\omega]_{\times}$ with $R(0) = I$, and its solution is $R(t) = \exp(t[\omega]_{\times})$, the exponential map. Why does $\exp$ land on the group for every t – what does it do that adding a small matrix δR did not – and predict which rotation $\exp$ returns for a tangent vector of length π .

Setup. The exponential map takes an element of $\mathfrak{m}$ to the group, $\exp : \mathfrak{m} \rightarrow \mathcal{M}$, and $\log$ is its inverse. In the plane $\exp(\theta J) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ with $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. In three dimensions $\exp([\omega]_{\times})$ is defined by the power series

$$\exp([\omega]_{\times}) = \sum_{k=0}^{\infty} \frac{1}{k!} [\omega]_{\times}^k = I + [\omega]_{\times} + \frac{1}{2} [\omega]_{\times}^2 + \frac{1}{6} [\omega]_{\times}^3 + \dots, \quad (3)$$

whose closed form is Rodrigues' formula. The capitalized $\text{Exp} : \mathbb{R}^m \rightarrow \mathcal{M}$ composes the hat with $\exp$ and sends a Cartesian vector τ straight to the group; Log is its inverse.

Facts to reason with.

- $\exp(0) = I$, $\exp(-v) = \exp(v)^{-1}$, and $\exp((s+t)v) = \exp(sv) \exp(tv)$.
- On Equation 1 the robot's heading after turning at rate ω for a time dt is $R \exp(\omega dt J)$.
- Only a few powers of $[\omega]_{\times}$ are nonzero, so the series in Equation 3 collapses to an ordinary 3×3 matrix.
- The closed form of $\log$ divides by $\sin \theta$, and the Jacobians divide by powers of θ , so the arithmetic needs care near a full turn.

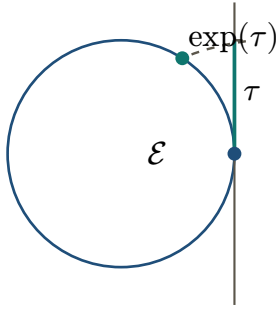


Figure 2: The tangent line at the identity of the circle, a tangent vector on it, and the point the exponential map sends that vector to. Inspect which point of the circle the far end reaches, and what must be true of that point for the result to be a rotation.

4. Adding and subtracting on the robot's pose

Question: Given $\exp$ and $\log$, define $X \oplus \tau = X \circ \exp(\tau)$ and $\tau = Y \ominus X = \text{Log}(X^{-1} \circ Y)$. Why do these two operations play the roles of $+$ and $-$ for the estimator, and why must a correction and a residual both be expressed in the tangent space rather than as differences of group elements?

Setup. The right-plus and right-minus operators combine a group element with a vector in its tangent space. The vector $\tau \in \mathbb{R}^m$ is the increment and X is the base point. Reversing the order gives the left versions, $\tau \oplus X$ and $X \ominus \tau$, which live in the tangent space at the identity.

Facts to reason with.

- $X \oplus 0 = X$, and $X \oplus \tau$ is a group element for every τ .
- The estimator's correction step is $\hat{X} \leftarrow \hat{X} \oplus \delta X$, and its residual is $z = y \ominus \hat{y}$; both δX and z are ordinary vectors.
- The order of the operands matters because the group is not commutative; the convention here is right-plus and right-minus throughout.
- $Y \ominus X$ and $X^{-1} \circ Y$ carry the same information, one as a vector and one as a group element.

5. The same increment in two frames

Question: The same small motion can be written as a local increment at X or as an increment at the identity: $X \oplus \tau = \tau' \oplus X$ for some τ' . The map between τ and τ' is linear and is called the adjoint. Why must such a map exist – what makes the two increments differ – and why does an estimator need it when it composes motions or moves a covariance between the body frame and the world frame?

Setup. For a group element X the adjoint is the map $\text{Ad}_X : \mathbb{R}^m \rightarrow \mathbb{R}^m$ defined by $\tau' = \text{Ad}_X \tau$, and it can be written as a matrix $\mathbf{Ad}_X$. It is the linearization of the conjugation $Y \mapsto X \circ Y \circ X^{-1}$ at the identity.

Facts to reason with.

- Ad_X is linear in τ , and $\text{Ad}_{XY} = \text{Ad}_X \text{Ad}_Y$.
- For a planar rotation the adjoint is the identity; for a 3D rotation it rotates the angular velocity vector; for a pose it mixes rotation and translation.
- A local perturbation of X and the global perturbation that produces the same Y are related by the adjoint, and so are their covariances.
- Composing two motions expressed in different frames requires moving one increment into the frame of the other.

6. Derivatives that obey a chain rule

Question: For a function $f : \mathcal{M} \rightarrow \mathcal{N}$ between manifolds, the ordinary derivative has no meaning because $f(X + h) - f(X)$ does not. Using $\oplus$ and $\ominus$, the right Jacobian is defined by a limit. Why is the result an ordinary matrix, what does it map between, and why does it satisfy a chain rule that lets every Jacobian be assembled from a few elementary blocks?

Setup. The right Jacobian of f at X is

$$(df(X))/(dX) = \lim_{\tau \rightarrow 0} \frac{f(X \oplus \tau) \ominus f(X)}{\tau} \in \mathbb{R}^{n \times m}, \quad (4)$$

and it maps a tangent vector at X to a tangent vector at $f(X)$. The elementary blocks are the Jacobians of $\exp$, of $\log$, of inversion, of composition, and of the group action; the right Jacobian of $\exp$ is written $J_{r(\tau)}$. The chain rule composes them.

Facts to reason with.

- Perturbations compose through the Jacobian: $f(X \oplus \tau) \approx f(X) \oplus J\tau$ for small τ .
- The right Jacobian of $\exp$ has a closed form built from $\sin \theta$, $\cos \theta$, and powers of θ .
- The chain rule holds: the Jacobian of a composition is the product of the Jacobians of its parts.
- The blocks are generic: they hold for every group in the family of Equation 1, not one at a time.

7. Where the covariance lives

Question: A Gaussian needs a mean and a covariance, but the naive $\Sigma = \mathbb{E}\left[(X - \bar{X})(X - \bar{X})^T\right]$ is meaningless because $X - \bar{X}$ is not defined. Where must the mean and the covariance live for the distribution to be well defined, and how does a nonlinear map propagate the covariance?

Setup. With $\oplus$ and $\ominus$ the perturbation of X is a vector, $X = \bar{X} \oplus \tau$ with $\tau \sim \mathcal{N}(0, \Sigma)$ and $\Sigma \in \mathbb{R}^{m \times m}$. The covariance can be expressed in the tangent space at the mean (local) or at the identity (global); the two are related by the adjoint.

Facts to reason with.

- m is the number of degrees of freedom of the state, so Σ has the size of a well-defined Gaussian even though the state does not.
- For a function f , the covariance propagates as $\Sigma_Y \approx J \Sigma_X J^T$ with J the right Jacobian of Equation 4.
- The ellipse that Σ describes sits in the tangent plane and wraps onto the manifold.
- On Equation 1 the pose covariance is 3×3 in the plane.

8. The estimator, written on the manifold

Question: Take the standard error-state Kalman filter. Its prediction and correction steps use ordinary $+$ and $-$ and a Kalman gain. Replacing those by $\oplus$ and $\ominus$, and computing the Jacobians with the manifold blocks, what do the two steps become – and why does the innovation stay an ordinary vector even though the state does not?

Setup. The state is $X \in \text{SE}(2)$ with error state $\delta X \in \mathbb{R}^3$ and covariance P . The prediction applies the control u as $X \leftarrow X \oplus u$; the correction forms the innovation $z = y \ominus \hat{y}$, the gain $K = PH^T(HPH^T + N)^{-1}$, and updates $X \leftarrow X \oplus (Kz)$.

Facts to reason with.

- The gain's formula is unchanged from the ordinary filter; only H is now a manifold Jacobian.
- A landmark-based smoother writes every residual as a $\ominus$ between a measurement and a prediction, and linearizes each with the blocks of Equation 4.
- The same recipe serves a pose, a rotation, or a direction: the group changes, the operators do not.

9. When the state is more than one group

Question: The estimator now also estimates an unknown sensor bias, so its state is a pose together with a vector: $\mathcal{X} = (X, c)$ with $X \in \text{SE}(2)$ and $c \in \mathbb{R}^2$. What must $\oplus$ and $\ominus$ do to such a state, what does the Jacobian of Equation 4 become, and why does the estimator still work when the two blocks have nothing to do with each other?

Setup. A composite manifold is a tuple of independent manifolds, $\mathcal{M} = (\mathcal{M}_1, \dots, \mathcal{M}_M)$, with the identity, inverse, composition, and $\oplus/\ominus$ acting block by block. On Equation 1 the state $\mathcal{X} = (X, c)$ pairs the pose with a bias; the measurement depends on both, so its Jacobian gains a column for the bias.

Facts to reason with.

- The blocks do not interact: composing (X_1, c_1) with (X_2, c_2) composes each coordinate on its own.
- The exponential of a composite is the tuple of the blocks' exponentials.
- A measurement residual is still a single $\ominus$ on the composite state.
- The self-calibrating smoother of the paper estimates the bias and the trajectory together.

10. Sources and further reading

This question map follows the selection of material in Joan Solà, Jeremie Deray, and Dinesh Atchuthan, **A micro Lie theory for state estimation in robotics**, arXiv:1812.01537[°], which the reader may consult for the full derivations and the reference tables of the common groups. No other external source has been consulted, and the note itself makes no claim about prior work, attribution, or the history of any method. The expected answers to the nine questions are kept in *summary.md* rather than in the document.